Part 1: Using both the theoretical and numerical methods, compute the off-axis compliance (or stiffness) matrix components for the composite described below.

Table 1: Material properties and fiber orientation

θ	E_1 (GPa)	$E_2 = E_3$ (GPa)	G_{12} (GPa)	ν_{12}	ν_{23}
T°	E	F	G	H	I

It is noted that

$$[\varepsilon'] = [\bar{S}][\sigma'] \tag{1}$$

where the off-axis compliance matrix is given by

$$[\bar{S}] = \begin{bmatrix} \bar{S}_{11} & \bar{S}_{12} & \bar{S}_{13} & 0 & 0 & 0 \\ \bar{S}_{12} & \bar{S}_{22} & \bar{S}_{23} & 0 & 0 & 0 \\ \bar{S}_{13} & \bar{S}_{23} & \bar{S}_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{S}_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \bar{S}_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & \bar{S}_{66} \end{bmatrix} \tag{2}$$

Hint: For the numerical method, in ABAQUS finite element software, apply a load in a basic direction and calculate the corresponding strain. Then repeat this process for the other directions (similar to an experimental procedure).

Note: Consider the parameters in Table 1 as functions of the last two digits of your student number (assume the student number has the form #####AB):

$$T = 45 + 4B,\ E = 120 + 5B,\ F = 10 + A,\ G = 5 + A,\ \nu_{12} = 0.25 + \frac{A}{100},\ \nu_{23} = 0.30 + \frac{A}{50}.$$

Solution

Theoretical method

According to student number, AB = 84. So,

$$\boxed{\theta = 61^\circ,\ E_1 = 140\,\text{GPa},\ E_2 = E_3 = 18\,\text{GPa},\ G_{12} = 13\,\text{GPa},\ \nu_{12} = 0.33,\ \nu_{23} = 0.46}$$

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & 0 & 0 \\ S_{12} & S_{22} & S_{23} & 0 & 0 & 0 \\ S_{13} & S_{23} & S_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & S_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{Bmatrix}$$

$$S_{11} = \frac{1}{E_1},\ S_{12} = -\frac{\nu_{12}}{E_1},\ S_{13} = -\frac{\nu_{13}}{E_1},$$

$$S_{22} = \frac{1}{E_2},\ S_{23} = -\frac{\nu_{23}}{E_2},\ S_{33} = \frac{1}{E_3},$$

$$S_{44} = \frac{1}{G_{23}},\ S_{55} = \frac{1}{G_{13}},\ S_{66} = \frac{1}{G_{12}}.$$

As $E_2 = E_3$, we assume it as a transversly isotropic and

$$\nu_{13} = \nu_{12}, \quad G_{13} = G_{12}$$

For other G_{23} parameters,

$$S = \begin{bmatrix} 7.14 & -2.36 & -2.36 & 0 & 0 & 0 \\ -2.36 & 55.56 & -25.56 & 0 & 0 & 0 \\ -2.36 & -25.56 & 55.56 & 0 & 0 & 0 \\ 0 & 0 & 0 & 162.34 & 0 & 0 \\ 0 & 0 & 0 & 0 & 76.92 & 0 \\ 0 & 0 & 0 & 0 & 0 & 76.92 \end{bmatrix} (\mathrm{TPa})^{-1}$$

for $\theta = 61°$, m $= cos\theta$, and n $= sin\theta$,

$$\bar{S} = T^T S T$$

$$= \begin{bmatrix} m^2 & n^2 & 0 & 0 & 0 & -mn \\ n^2 & m^2 & 0 & 0 & 0 & mn \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & m & n & 0 \\ 0 & 0 & 0 & -n & m & 0 \\ 2mn & -2mn & 0 & 0 & 0 & m^2-n^2 \end{bmatrix} S \begin{bmatrix} m^2 & n^2 & 0 & 0 & 0 & 2mn \\ n^2 & m^2 & 0 & 0 & 0 & -2mn \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & m & -n & 0 \\ 0 & 0 & 0 & n & m & 0 \\ -mn & mn & 0 & 0 & 0 & m^2-n^2 \end{bmatrix}$$

$$\bar{S} \approx \begin{bmatrix} 45.89 & -4.07 & -20.11 & 0 & 0 & -18.40 \\ -4.07 & 20.23 & -7.81 & 0 & 0 & -22.67 \\ -20.11 & -7.81 & 55.56 & 0 & 0 & 19.67 \\ 0 & 0 & 0 & 97.00 & -36.22 & 0 \\ 0 & 0 & 0 & -36.22 & 142.26 & 0 \\ -18.40 & -22.67 & 19.67 & 0 & 0 & 70.09 \end{bmatrix} (\mathrm{TPa})^{-1}$$

Numerical method

As hinted in the question, we use ABAQUS for the numerical solution.

Part

For the model, we create a $1 \times 1 \times 1\,\mathrm{mm}^3$ cube.

Figure 1: Cube part geometry

Property

We use the values calculated for the composite material in the previous part:

$$E_1 = 140\,\text{GPa},\ E_2 = E_3 = 18\,\text{GPa},\ G_{12} = G_{13} = 13\,\text{GPa},\ G_{23} = 6.16\,\text{GPa},$$
$$\nu_{12} = \nu_{13} = 0.33,\ \nu_{23} = 0.46$$

These are entered in the property module as **Engineering Constants**.

Figure 2: Material Properties

Our material has a $\theta = 61°$ orientation around the z-axis. We define a datum to set a new coordinate system at the origin by a $61°$ rotation around the z-axis.

Figure 3: Create Datum

Then, we assign the orientation to the material.

Figure 4: Assign orientation to material

Step
Create a Static, General step after the initial step.

Figure 5: Creating a static, general step

Load

The important part of this process is the load module. To find the compliance matrix as in experimental methods, we should apply six separate loading cases to our part. In the following table, we define each loading.

Table 2: Load cases and corresponding surface tractions (MPa) and surface directions.

Case	**Target stress**	**X+**	**X-**	**Y+**	**Y-**	**Z+**	**Z-**
1	$\sigma_{11} = 1$	(1,0,0)	(-1,0,0)	-	-	-	-
2	$\sigma_{22} = 1$	-	-	(0,1,0)	(0,-1,0)	-	-
3	$\sigma_{33} = 1$	-	-	-	-	(0,0,1)	(0,0,-1)
4	$\sigma_{23} = 1$	-	-	(0,0,1)	(0,0,-1)	(0,1,0)	(0,-1,0)
5	$\sigma_{13} = 1$	(0,0,1)	(0,0,-1)	-	-	(1,0,0)	(-1,0,0)
6	$\sigma_{12} = 1$	(0,1,0)	(0,-1,0)	(1,0,0)	(-1,0,0)	-	-

Figure 6: Defining normal stress in the Load module

Figure 7: Defining shear stress in the Load module

For the boundary condition, we simply fix the corner at the part's origin.

Mesh

We use C3D8 elements and mesh the entire part with a single element to reduce calculations, because there is no difference in the results based on our modelling method.

Figure 8: Single-element C3D8 mesh

Job

We save six copies of the model, each with a specific loading case applied, and create a corresponding job for each.

Figure 9: Taking strains as outputs

Results

To obtain the correct outputs, the results must first be transformed. Initially, the results are expressed in the local material orientation. From the Tools menu, we create a coordinate system that is identical to the global coordinate system. Then, in the Results module, we go to Outputs and transform the results to the global coordinate system. Finally, from the Report Field Output menu, we select the strains and save the data in a .txt file. In the end, we have six .txt files containing the strains for each loading case.

Table 3: Strain results for each unit-stress loading case (stress applied = 1 MPa).

	ϵ_{11}'	ϵ_{22}'	ϵ_{33}'	γ_{23}'	γ_{13}'	γ_{12}'
$\sigma_{11}' = 1$	45.8865E-06	-4.06708E-06	-20.1030E-06	53.2572E-21	54.4219E-21	-18.3912E-06
$\sigma_{22}' = 1$	-4.06708E-06	20.2318E-06	-7.80973E-06	39.5988E-21	47.0103E-21	-22.6651E-06
$\sigma_{33}' = 1$	-20.1030E-06	-7.80973E-06	55.5556E-06	-60.5629E-21	-42.2458E-21	19.6734E-06
$\sigma_{23}' = 1$	142.725E-21	42.7752E-21	-115.196E-21	96.9990E-06	-36.2179E-06	-170.571E-21
$\sigma_{13}' = 1$	36.8459E-21	36.8459E-21	-37.2695E-21	-36.2179E-06	142.262E-06	-178.512E-21
$\sigma_{12}' = 1$	-18.3912E-06	-22.6651E-06	19.6734E-06	-107.997E-21	-166.442E-21	70.0833E-06

For case 1:

$$\begin{Bmatrix} \bar{S}_{11} \\ \bar{S}_{21} \\ \bar{S}_{31} \\ \bar{S}_{41} \\ \bar{S}_{51} \\ \bar{S}_{61} \end{Bmatrix} \approx \begin{Bmatrix} 45.89 \\ -4.07 \\ -20.10 \\ 0 \\ 0 \\ -18.39 \end{Bmatrix} (\mathrm{TPa})^{-1}$$

Similarly, we can obtain the other $\bar{S}$ indices, and the compliance matrix becomes

$$\bar{S} \approx \begin{bmatrix} 45.89 & -4.07 & -20.10 & 0 & 0 & -18.39 \\ -4.07 & 20.23 & -7.81 & 0 & 0 & -22.67 \\ -20.10 & -7.81 & 55.56 & 0 & 0 & 19.67 \\ 0 & 0 & 0 & 97.00 & -36.22 & 0 \\ 0 & 0 & 0 & -36.22 & 142.26 & 0 \\ -18.39 & -22.67 & 19.67 & 0 & 0 & 70.08 \end{bmatrix} (\mathrm{TPa})^{-1}$$

The matrices obtained from both methods have the same values, with very little error.

Part 2: Use UMAT subroutine and find distribution of three invariants of the stress tensor in a circular plate under radial stress uniformly distributed around its circumference. (Select and report all necessary data)

Solution

Theoretical method

At any point within the circular composite plate, the state of stress is defined by a second-order symmetric tensor, $\boldsymbol{\sigma}$. In a general 3D Cartesian coordinate system (x, y, z corresponding to directions $1, 2, 3$), it is expressed as:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \tau_{12} & \tau_{13} \\ \tau_{12} & \sigma_{22} & \tau_{23} \\ \tau_{13} & \tau_{23} & \sigma_{33} \end{bmatrix}$$

Because the stress tensor is symmetric ($\tau_{ij} = \tau_{ji}$), it possesses six independent components. In Abaqus, these components are passed into the UMAT subroutine as a 1D vector (STRESS) whose indexing depends on the element type.

Stress invariants are quantities that do not change with a rotation of the coordinate system. Even though your composite material has highly directional properties (causing the stresses σ_{11} and σ_{22} to align strongly with the stiff fiber direction), the mathematical definitions of the invariants remain entirely objective.

They are derived from the characteristic equation of the stress tensor, obtained by finding the eigenvalues (principal stresses, σ):

$$\det(\boldsymbol{\sigma} - \sigma \mathbf{I}) = 0$$

Expanding this determinant yields a cubic polynomial equation:

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$$

Where I_1, I_2, and I_3 are the three principal invariants of the stress tensor.

The First Invariant (I_1)

The first invariant represents the trace of the stress tensor. Physical-mechanically, it is directly proportional to the hydrostatic (volumetric) stress acting on the material point.

$$I_1 = \mathrm{tr}(\boldsymbol{\sigma}) = \sigma_{11} + \sigma_{22} + \sigma_{33}$$

The Second Invariant (I_2)

The second invariant is related to the magnitude of the shear stresses and the distortional energy stored in the material. It can be computed using the sum of the principal minors of the matrix:

$$I_2 = \sigma_{11}\sigma_{22} + \sigma_{22}\sigma_{33} + \sigma_{33}\sigma_{11} - \tau_{12}^2 - \tau_{23}^2 - \tau_{13}^2$$

Alternatively, it can be written compactly using tensor notation as:

$$I_2 = \frac{1}{2}\left[(\mathrm{tr}\boldsymbol{\sigma})^2 - \mathrm{tr}(\boldsymbol{\sigma}^2)\right]$$

The Third Invariant (I_3)

The third invariant is the determinant of the stress tensor matrix. It represents the product of the three principal stresses ($\sigma_I \cdot \sigma_{II} \cdot \sigma_{III}$).

$$I_3 = \det(\boldsymbol{\sigma}) = \sigma_{11}\sigma_{22}\sigma_{33} + 2\tau_{12}\tau_{23}\tau_{13} - \sigma_{11}\tau_{23}^2 - \sigma_{22}\tau_{13}^2 - \sigma_{33}\tau_{12}^2$$

Since the problem statement describes a circular plate, it is standard practice to model this geometry using Plane Stress finite elements. By definition, plane stress assumes that all out-of-plane stress components acting normal to the plate's surface are zero:

$$\sigma_{33} = 0, \quad \tau_{13} = 0, \quad \tau_{23} = 0$$

Applying these zero-value constraints simplifies the invariant equations computed inside your UMAT subroutine down to:

– First Invariant:

$$I_1 = \sigma_{11} + \sigma_{22}$$

– Second Invariant:

$$I_2 = \sigma_{11}\sigma_{22} - \tau_{12}^2$$

– Third Invariant:

$$I_3 = 0$$

While I_3 will mathematically remain zero everywhere due to the plane stress state, the distributions of I_1 and I_2 across the circular plate will not be uniform. Because fibers ($E_1 = 140$ GPa) carry significantly more load than the matrix ($E_2 = 18$ GPa), σ_{11} will be much larger than σ_{22} near the center, generating distinct contour variations for I_1 and I_2 across the plate's geometry.

Numerical method

Now we want to implement UMAT subroutine in ABAQUS. The parameters decided in this implementation are listed in following table..

Table 4: Design Parameters and Modeling Decisions for the Circular Composite Plate Analysis

Parameter	**Symbol**	**Selected Value**
Plate Radius	r	100 mm
Plate Thickness	t	2 mm
Outer Radial Traction	σ_0	50 MPa
Space Type	—	2D Planar Shell
Element Type	—	CPS4
State Variables (SDVs)	—	3
Center Constraint	—	$U_x = U_y = 0$

Part

A 2D planar deformable shell part is sketched as a circle with a 100 mm radius to establish the geometry of the plate.

Figure 10: Circular 2D plate part

And Partitioning was performed for better meshing and selection of the midpoint for boundary conditions.

Figure 11: Creat partitions for better meshing

Property

A user material section with three solution-dependent variables is created and assigned to the plate with an explicit thickness of 2 mm using the From section option.

Figure 12: Use UMAT for material properties

Assembly

The circular plate geometry is instanced into the global coordinate system to make it available for analysis steps and loading.

Figure 13: Creat assembly

Step

A general static analysis step is configured with an initial increment size of 0.1 to apply the radial load incrementally

Figure 14: General static step

Load Module

The plate center is pinned with a displacement constraint ($U_x = U_y = 0$) to eliminate rigid body motion, and a negative pressure load of -50 MPa is applied along the outer perimeter to generate uniform radial tension.

Figure 15: Define a negative pressure load of -50 MPa

Figure 16: Pin the center of the plate

Mesh Module

The circular surface is seeded with a global size of 5.0 and discretized using linear, 4-node plane stress elements (CPS4).

Figure 17: Creat CPS4 mesh

Job

A simulation job is created, linked directly to the external `umat_composite.f` Fortran subroutine script, and submitted to the Abaqus solver engine.

Figure 17: Use the umat wrritten file for solve

Result

The completed output file is opened to plot the field contours of the state variables, confirming uniform distributions of 100 MPa for SDV1 (I_1), 2500 MPa^2 for SDV2 (I_2), and a constant zero for SDV3 (I_3).

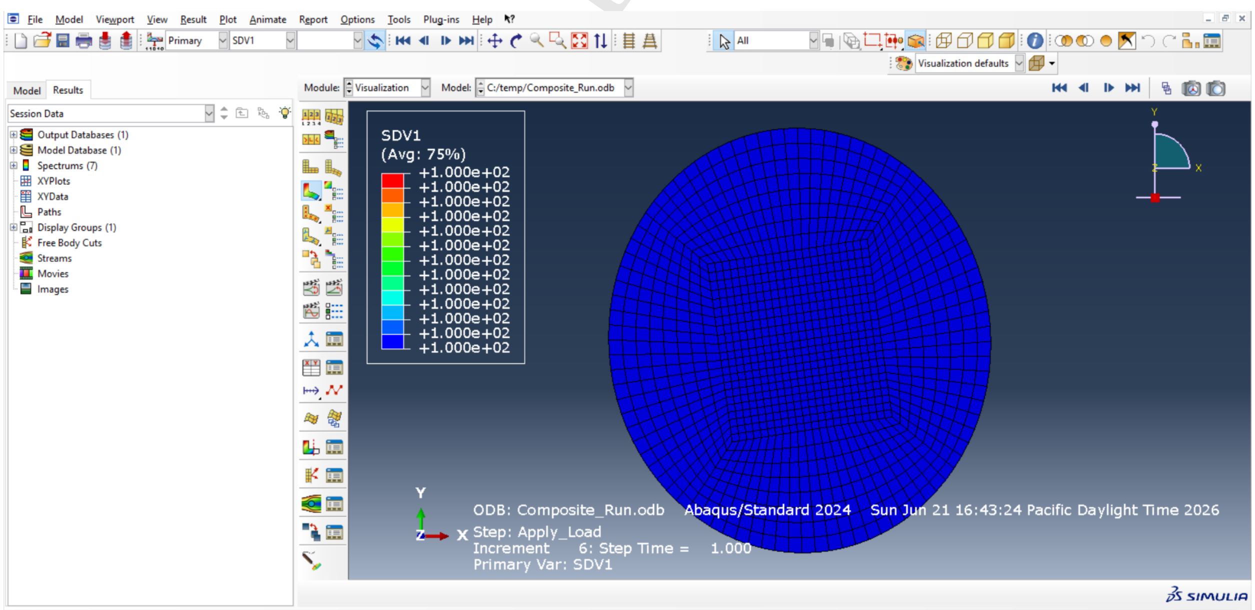

Figure 18: SDV1 value

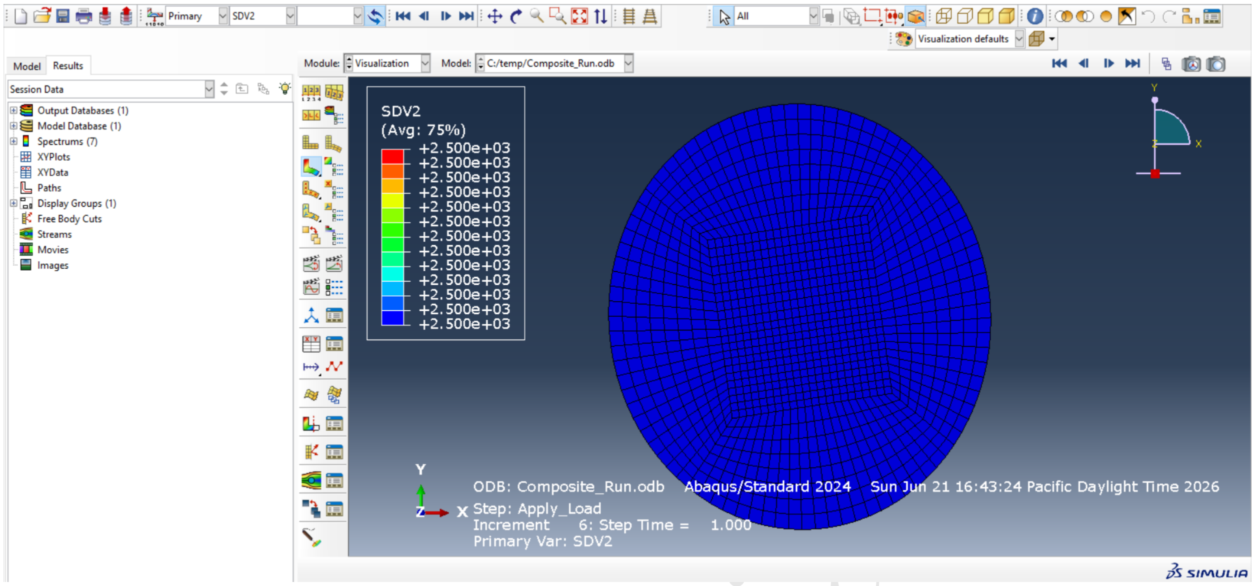

Figure 19: SDV2 value

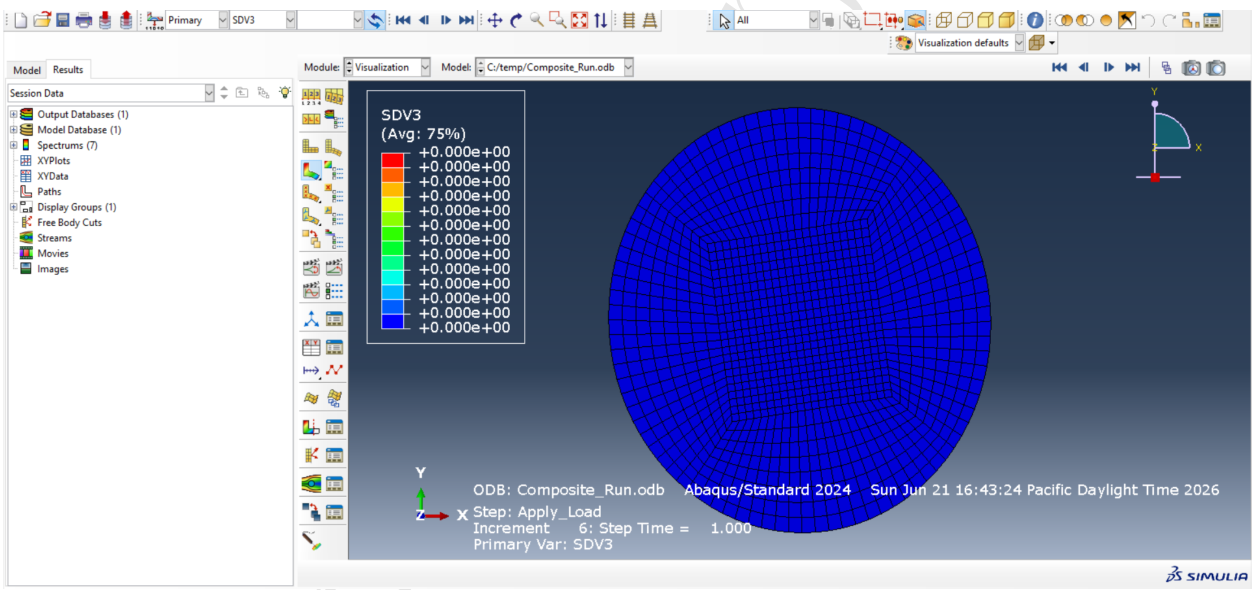

Figure 20: SDV3 value

Table 5: Abaqus UMAT Simulation Output Values

Output Parameter	**UMAT Variable**	**Simulation Result**
First Stress Invariant	SDV1	100 MPa
Second Stress Invariant	SDV2	2500 MPa^2
Third Stress Invariant	SDV3	0 MPa^3